

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: CSA14

COURSE NAME: COMPILER DESIGN

COURSE OUTCOME COVERED

CO5: Develop code optimization and Code Generation using DAG representation with data flow analysis to produce optimized target code.(BL3)

Assessment Tool Weightage:10%

Duration :60 Mins
On Class
Total Marks:50

QUESTIONS: 50

Annexure A

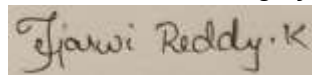
MOCK INTERVIEW

Code of Conduct:

I Tejaswi Reddy K (192371036) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Q. No	Technical Knowledge(25)	Problem Solving(25)	Analytical Thinking(15)	Communication(15)	Confidence (10)	Response to Questions(10)	Total Score (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
4	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
5	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
	OVERALL RUBRICS VALUE						
	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 25	/ 25	/ 15	/ 15	/ 10	/ 10	/ 100

Annexure A

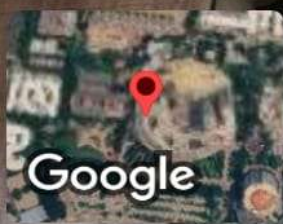
MOCK INTERVIEW

- Q1. What is code generation in a compiler, and why is it considered the final phase of compilation?
- Q2. What is the difference between code generation and code optimization?
- Q3. What are the primary objectives of code optimization?
- Q4. What are the principal sources of optimization in a compiler?
- Q5. How does machine-independent optimization differ from machine-dependent optimization?
- Q6. What factors influence the quality of generated target code?
- Q7. Why is reducing execution time an important goal of optimization?
- Q8. What trade-offs exist between code size and execution speed during optimization?
- Q9. What is peephole optimization?
- Q10. Why is peephole optimization considered a machine-dependent optimization?
- Q11. What are the common techniques used in peephole optimization?
- Q12. How does redundant instruction elimination improve target code?
- Q13. Explain algebraic simplification with an example.
- Q14. What is strength reduction, and when is it applied?
- Q15. How does jump optimization reduce unnecessary control transfers?
- Q16. How would you optimize the instruction sequence `MOV R1,R2` followed by `MOV R2,R1`?
- Q17. What is a basic block?
- Q18. How do you identify the leaders of basic blocks?
- Q19. Why must a basic block have only one entry and one exit?
- Q20. How do you construct basic blocks from three-address code?
- Q21. What is the purpose of optimizing a basic block?
- Q22. How does common subexpression elimination work inside a basic block?
- Q23. What is copy propagation, and how does it simplify code?

- Q24. How does dead code elimination improve basic block optimization?
- Q25. What is a flow graph?
- Q26. How are basic blocks represented in a flow graph?
- Q27. Why are edges important in a control flow graph?
- Q28. How can a compiler identify unreachable basic blocks?
- Q29. What is global data flow analysis?
- Q30. How does data flow analysis differ from local optimization?
- Q31. What is reaching definitions analysis?
- Q32. What is liveness analysis, and why is it useful?
- Q33. What are available expressions in data flow analysis?
- Q34. How does loop-invariant code motion improve performance?
- Q35. What are the major issues involved in designing a code generator?
- Q36. How does the target machine architecture affect code generation?
- Q37. What role do CPU registers play in efficient code generation?
- Q38. Why is register allocation an important part of code generation?
- Q39. What is register spilling, and when does it occur?
- Q40. How does instruction selection influence generated machine code?
- Q41. What is runtime storage management?
- Q42. What is an activation record, and what information does it contain?
- Q43. What is the difference between stack memory and heap memory?
- Q44. How are local variables managed during function execution?
- Q45. What is next-use information?
- Q46. How does next-use information help in register allocation?
- Q47. What is a Directed Acyclic Graph (DAG) representation of a basic block?
- Q48. How does a DAG help eliminate common subexpressions?
- Q49. How would you generate target code from a DAG representation?
- Q50. If you were designing a compiler for a processor with only two registers, what optimization strategies would you use to generate efficient code?



GPS Map Camera



Mevalurkuppam, Tamil Nadu, India

EEE Department, Mevalurkuppam, Tamil Nadu 602105, India

Lat 13.026394° Long 80.016063°

27/08/2026 09:03 AM GMT +05:30

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	15	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism
Response to Questions	10	Responds accurately, promptly, and elaborates with relevant examples	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriately